

Linear_Regression

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202210183

```
[1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, OneHotEncoder

companies = pd.read_csv("1000_Companies.csv")
X = companies.iloc[:, :-1]
Y = companies.iloc[:, 4]

labelencoder = LabelEncoder()
X["State"] = labelencoder.fit_transform(X["State"])

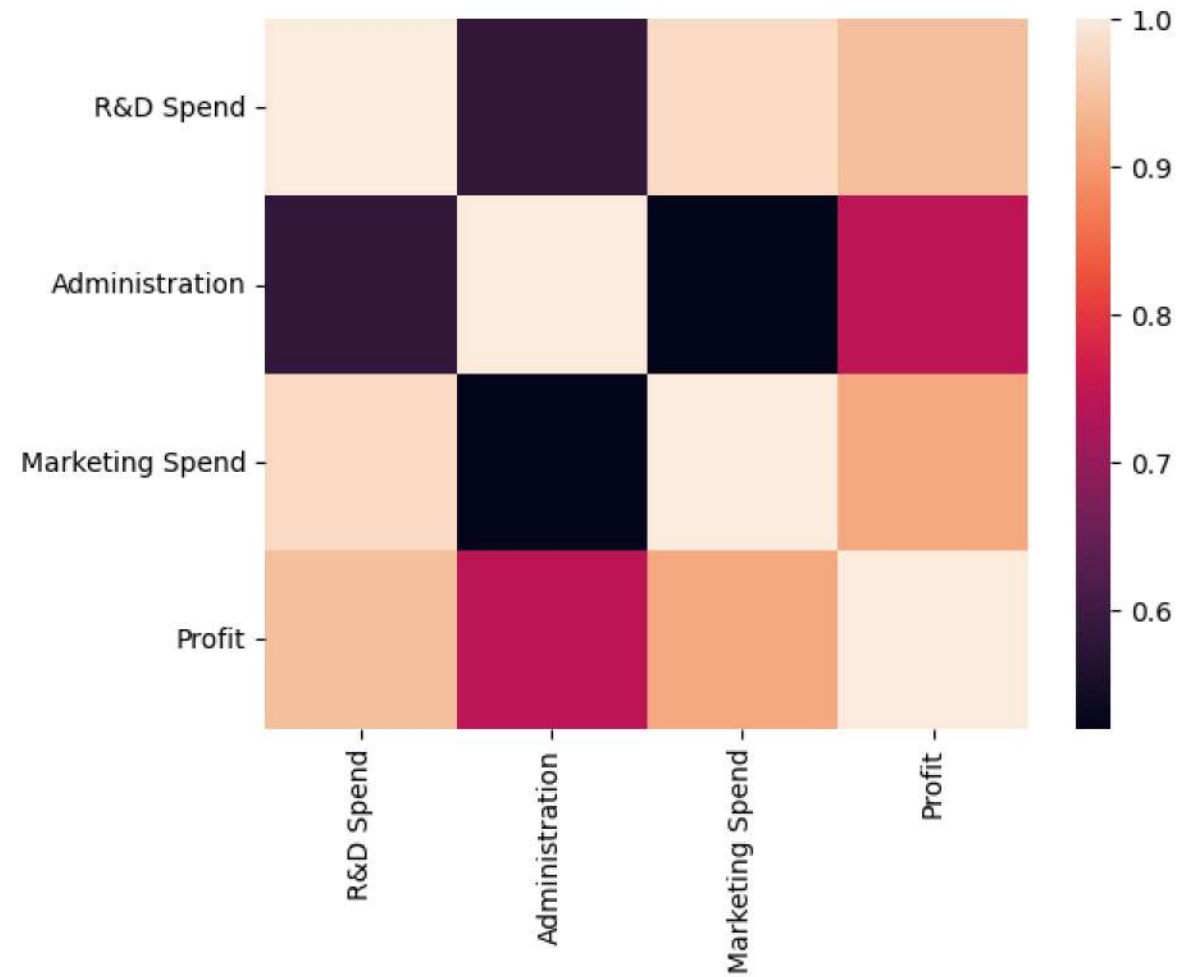
companies.head()
```

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[1]:
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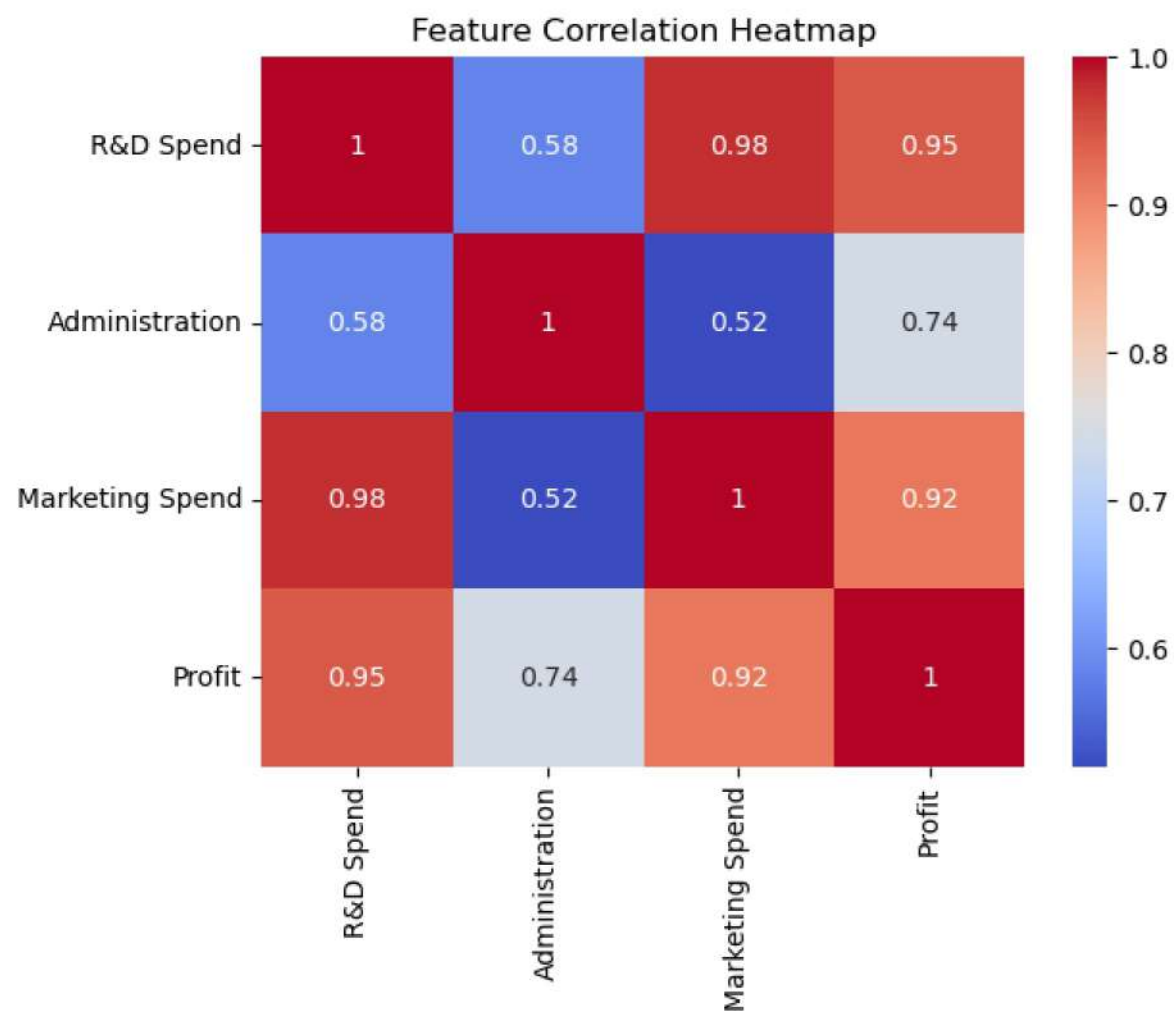
	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

```
[2]: sns.heatmap(companies.drop('State', axis=1) .corr())
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[2]: <Axes: >
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[3]: sns.heatmap(companies.drop('State', axis=1).corr(), annot=True, cmap='coolwarm')  
plt.title("Feature Correlation Heatmap")  
plt.show()
```



```
[4]: from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import r2_score

      X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.2, random_state = 0)

      model_fit = LinearRegression()
      model_fit.fit(X_train, y_train)

      y_pred = model_fit.predict(X_test)
      print(f"The Coefficients : {model_fit.coef_}")
      print(f"The Intercept : {model_fit.intercept_}")

      print(r2_score(y_test, y_pred))
```

```
The Coefficients : [ 5.27572958e-01  8.44490415e-01  1.06639226e-01
-3.54869857e+02]
The Intercept : -51144.726425721834
0.9112595959622045
```

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[5]: from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import r2_score

      X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.25, random_state = 0)

      model_fit = LinearRegression()
      model_fit.fit(X_train, y_train)

      y_pred = model_fit.predict(X_test)
      print(f"The Coefficients : {model_fit.coef_}")
      print(f"The Intercept : {model_fit.intercept_}")

      print(r2_score(y_test, y_pred))
```

```
The Coefficients : [ 5.26512651e-01  9.03566864e-01  1.02337389e-01
-2.38604774e+02]
The Intercept : -57545.85107047536
0.9209548650003682
```

```
[6]: from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import r2_score

      X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.3, random_state = 0)

      model_fit = LinearRegression()
      model_fit.fit(X_train, y_train)

      y_pred = model_fit.predict(X_test)
      print(f"The Coefficients : {model_fit.coef_}")
      print(f"The Intercept : {model_fit.intercept_}")

      print(r2_score(y_test, y_pred))
```

The Coefficients : [5.28077058e-01 9.78263293e-01 9.70601727e-02
-2.78834327e+02]
The Intercept : -65740.27690882178
0.931091029778443

```
[7]: from sklearn.model_selection import train_test_split
      from sklearn.linear_model import LinearRegression
      from sklearn.metrics import r2_score

      X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.5, random_state = 0)

      model_fit = LinearRegression()
      model_fit.fit(X_train, y_train)

      y_pred = model_fit.predict(X_test)
      print(f"The Coefficients : {model_fit.coef_}")
      print(f"The Intercept : {model_fit.intercept_}")

      print(r2_score(y_test, y_pred))
```

The Coefficients : [1.89248431e-01 1.36148420e+00 2.44564040e-01
-4.00351514e+02]
The Intercept : -118112.47172057483
0.9262969754760986